Vishnu Kiran Dabbara

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LinkedIn | **Portfolio**

Summary

Senior .NET Full Stack Developer with around **9 years** of experience designing and delivering scalable, enterprise-grade applications across transportation, banking, and automation domains. Proficient in C#, .NET Core, ASP.NET Web API, and Microservices architecture, with strong expertise in Microsoft Azure (App Services, Functions, Service Bus, Azure AD). Experienced in building real-time, event-driven systems including ISO 8583-based banking solutions, middleware platforms, and RPA tools with bot orchestration. Skilled in performance optimization, secure application design (OAuth 2.0, JWT, RBAC), and CI/CD using Azure DevOps, Docker, and Kubernetes. Adept at Agile collaboration and delivering high-quality, reliable, and scalable solutions.

Skills

- **Languages & Frameworks:** C#, .NET Core, ASP.NET Web API, ASP.NET MVC, WPF, Windows Workflow Foundation (WF), WCF, Angular, JavaScript, Node.js
- **Backend Development:** REST APIs, Microservices, Entity Framework Core, LINQ, OOP, Distributed Systems
- **Database Management:** SQL Server, PostgreSQL, MongoDB, Redis
- **Cloud Technologies:** Microsoft Azure (App Services, Functions, API Management, Service Bus, Azure AD, Application Insights), AWS (EC2, S3, RDS, Lambda).
- **DevOps & CI/CD:** Azure DevOps, Docker, Kubernetes (AKS), Terraform.
- **Testing & Security:** NUnit, xUnit, OAuth 2.0, JWT, RBAC.
- **Version:** Git, JIRA, Agile (Scrum).

Experience

NYSDOT, Albany, NY **July 2023 - Current**

Sr Dot.NET Developer

- Designed and developed **scalable transportation applications** using **.NET Core, C#, ASP.NET Web API, and Microservices architecture**, supporting real-time traffic, asset, and infrastructure data processing.
- Built and deployed **cloud-native solutions on Microsoft Azure**, leveraging **Azure App Services, Azure Functions, and Azure API Management** to deliver highly available and secure APIs.
- Developed **event-driven data pipelines** using **Azure Service Bus and Azure Event Grid**, enabling real-time ingestion and processing of traffic sensors and roadway data.
- Designed and optimized **data integration workflows** using **Azure Data Factory and Azure Synapse Analytics**, handling large-scale transportation datasets for reporting and analytics.
- Implemented **secure authentication and authorization** using **Azure Active Directory (Azure AD), OAuth 2.0, and JWT**, ensuring compliance with state and federal security standards.
- Built interactive **Power BI dashboards** to visualize traffic trends, infrastructure performance, and maintenance KPIs for decision-making by stakeholders.
- Improved application performance using **Redis caching and asynchronous programming (async/await)**, reducing API response time by **40%**.
- Developed and optimized **SQL Server queries, stored procedures, indexing, and partitioning strategies**, improving performance for high-volume transportation data.
- Containerized applications using **Docker** and deployed to **Azure Kubernetes Service (AKS)**, enabling scalability and efficient resource utilization.
- Implemented **CI/CD pipelines using Azure DevOps (YAML pipelines)**, automating build, test, and deployment processes across environments.
- Integrated **GIS and spatial data services (ArcGIS APIs)** for mapping and geospatial analysis of transportation networks.
- Built robust **logging, monitoring, and alerting systems** using **Azure Application Insights and Log Analytics**, improving system reliability and incident response.
- Collaborated with **Scrum teams**, using **JIRA and Git**, ensuring timely delivery of features and enhancements.

Technologies: C#, .NET Core, ASP.NET Web API, Microservices, Azure (App Services, Functions, API Management, Service Bus, Event Grid, Data Factory, Synapse, Azure AD, Application Insights), SQL Server, Power BI, Redis, Docker, Kubernetes (AKS), OAuth 2.0, JWT, ArcGIS APIs, Git, Azure DevOps, CI/CD, JIRA.

Krisfo Infotech Pvt Ltd, Bengaluru, India **Oct 2020 - Sep 2022**

Senior Full Stack .Net Developer

- Designed and developed **enterprise-grade passbook printing applications** for banking clients including **State Bank of India, Canara Bank, and Indian Overseas Bank**, using **.NET Core, C#, and ASP.NET Web API** to enable secure, real-time transaction printing.
- Built and integrated **ISO 8583-based messaging systems** between passbook kiosks and Core Banking Systems (CBS), enabling seamless **transaction request/response processing via middleware**.
- Developed and maintained **middleware services** to handle communication between passbook printing machines, kiosks, and CBS systems, ensuring reliable message routing, validation, and transformation.
- Implemented **Remote Monitoring & Management System (RMMS)** to track kiosk health, uptime, and operational status, enabling proactive issue detection and reducing downtime.

- Engineered **real-time transaction processing workflows**, receiving transaction data from CBS and converting it into **printable formats for passbook machines**.
- Integrated **event-driven logging and audit trails** using **Apache Kafka and MongoDB**, ensuring compliance with **RBI, PCI DSS, and SOX regulations**.
- Improved performance by implementing **multi-threaded processing and background job scheduling using Hangfire**, reducing passbook update latency by **60%**.
- Secured applications using **AES-256 encryption, TLS 1.3, OAuth 2.0, and RBAC**, protecting sensitive financial data and ensuring secure communication across systems.
- Optimized database performance using **SQL Server columnstore indexing, partitioning, and query tuning**, improving transaction retrieval speed by **45%**.
- Migrated legacy passbook applications to **AWS cloud (ECS, RDS, CodePipeline)**, improving scalability, availability, and reducing infrastructure costs by **30%**.
- Automated workflow orchestration using **AWS Step Functions and AWS Lambda**, reducing manual intervention in reconciliation and printing processes by **70%**.

Technologies: C#, .NET Core, ASP.NET Web API, Entity Framework Core, SQL Server, ISO 8583, Apache Kafka, MongoDB, Hangfire, AWS (ECS, RDS, Lambda, Step Functions, CodePipeline), REST APIs, AES-256, TLS 1.3, OAuth 2.0, RBAC, Middleware Architecture, RMMS, Git, JIRA.

GET AI (formerly Ai-Bots Tech Private Limited), Bengaluru, India **Jan 2017 - Oct 2020**
Software Engineer

- Developed an enterprise-grade **RPA platform (Studio + Orchestrator)** using **.NET Core, C#, ASP.NET MVC, and WPF**, enabling end-to-end automation of business processes.
- Built a **drag-and-drop workflow designer (RPA Studio)** using **Windows Workflow Foundation (WF)**, allowing users to create and manage automation workflows without coding.
- Designed and developed an **Orchestrator module** to **schedule, monitor, and manage bots**, including job queues, triggers, and execution logs.
- Implemented **Bot scheduling and queue management system**, enabling unattended automation and improving operational efficiency.
- Developed reusable **custom activity libraries** using **C# and XAML** for automation across **Web, Desktop, and Citrix environments**.
- Integrated **REST and SOAP APIs** for seamless interaction with external systems like **banking, ERP, and CRM platforms**.
- Implemented **secure authentication and authorization** using **Azure Active Directory (Azure AD), OAuth 2.0, and JWT**, supporting role-based access control (RBAC).
- Designed and optimized **database schemas and queries** using **SQL Server and Entity Framework Core**, improving performance of bot execution tracking and logs.
- Built **real-time monitoring dashboards** for Bot execution status, failures, and performance metrics.
- Deployed scalable components using **Azure Virtual Machines and Azure Service Bus**, ensuring high availability of Bot execution.
- Implemented **centralized logging and exception handling** using **Serilog and Azure Application Insights** for better observability.
- Automated **CI/CD pipelines** using **Azure DevOps**, reducing deployment time and improving release reliability.
- Enhanced system performance using **asynchronous processing and caching mechanisms**, improving Bot execution speed and scalability.

Technologies: C#, .NET Core, ASP.NET MVC, WPF, XAML, Windows Workflow Foundation (WF), Entity Framework Core, SQL Server, REST APIs, SOAP, Azure (Azure AD, VM, Service Bus, Application Insights), Azure DevOps, CI/CD, OAuth 2.0, JWT, Serilog, Redis, Docker.

Education

Master Computer Science and Information Systems **Aug 2022 - Dec 2023**
Youngstown State University, United States
Bachelor of Technology Computer Science and Engineering **Jun 2013 - Jul 2017**
MIC College of Technology, India